DATA SCIENCE IN HEALTH CARE · FINAL PROJECT

Survival after pediatric stem cell transplantation

Does the stem-cell source matter, or is it the CD34+ cell dose?

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BACKGROUND

Three terms you need for this talk

01

Stem cell transplant (HSCT)

Replaces a child's diseased blood-forming system with healthy donor stem cells. Often the only cure for leukaemia, lymphoma or bone-marrow failure.

02

Stem-cell source

Bone marrow (BM): harvested from the donor's hip bone. **Peripheral blood (PB):** collected from the donor's blood after injections that mobilise stem cells.

03

CD34+ cell dose

CD34 marks stem and progenitor cells. The dose ($\times 10^6$ cells per kg of body weight) measures how much graft the child receives.

Why it matters: in our cohort **85 of 187 children (45%) died** during follow-up. Source and dose are choices the transplant team can influence.

THE PUZZLE

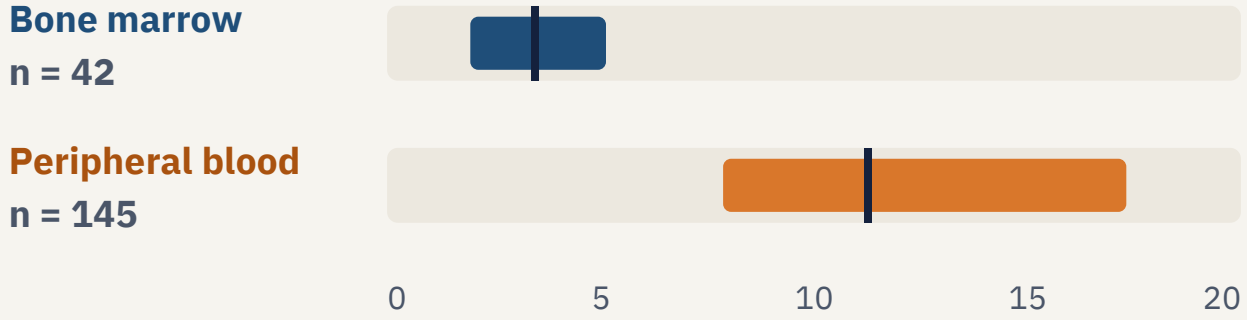
Source and dose travel together

3.3×

more CD34+ cells per kg in peripheral-blood grafts

Median 11.3 vs 3.4 ×10⁶/kg
Mann–Whitney p < 0.001

CD34+ dose by stem-cell source (bar = middle 50% of patients, tick = median)



×10⁶ CD34+ cells per kg · dose also falls with age (Spearman ρ = −0.46)

If PB patients survive longer, is it the source, or the dose that that comes with it?

RESEARCH QUESTION

Hypotheses with explicit tests

Is stem-cell source associated with overall survival in pediatric HSCT, and does the association remain after accounting for accounting for CD34+ dose and clinical factors?

#	Hypothesis	Test and statistic	Decision (two-sided, α = 0.05)
H1a	Stem-cell source is associated with survival (unadjusted)	Log-rank χ^2 test; Wald test in Cox Model 1	$p < 0.05$
H1b	The source effect weakens after adjusting for CD34+ dose	% change in the source log-HR from Model 1 to Model 2, with a bootstrap 95% CI of the change (1,000 resamples)	Change ≥ 10% and CI excludes 0
H2	Higher CD34+ dose means longer survival, after adjustment	Wald test of the CD34+ coefficient in Cox Model 3 (H_0: HR = 1)	HR < 1 and $p < 0.05$

DATA

187 children, one unrelated-donor transplant each

187

pediatric patients, 37 variables

85

deaths (45%); 102 alive at last follow-up

145 / 42

peripheral blood / bone marrow

676

median days observed (range 6–3,364)

Diagnosis



ALL 68 · **Chronic leukaemia** 45 · **AML** 33 · **Non-malignant** 32 · **Lymphoma** 9 (all 9 died)

Source: UCI ML Repository, Bone Marrow Transplant: Children (Sikora & Wróbel)

DATA PREPARATION

Every data issue got an explicit rule

Issue found	Decision
Missing values (at most 8.6%, CMV status)	Cox: every variable used is complete, so all 187 patients, no imputation. Random forest: median / most-frequent imputation inside the CV pipeline, fitted on training folds only.
“1,000,000” in recovery and GvHD times	A placeholder for “not observed”: converted to censoring or a competing event, never used as a time.
Post-transplant variables (relapse, GvHD, recovery)	Excluded as predictors: they occur after transplant and would leak the outcome.
2 records: platelet recovery dated after death	Impossible order: treated as “no recovery before death”.
Every non-engrafted patient died (5 neutrophil, 17 platelet)	Death modelled as a competing event in the engraftment analysis.

METHODS

Four steps, from description to prediction

1 · DESCRIBE

Kaplan–Meier

- Overall survival
- PB vs BM curves
- Log-rank test

2 · EXPLAIN

Cox models

- Models 1 → 2 → 3 → 3q
- Schoenfeld PH tests
- Bootstrap, sensitivity

3 · PREDICT

Survival forest

- RSF vs Cox
- 5 × 5-fold CV
- C-index, Brier score

4 · SECONDARY

Engraftment

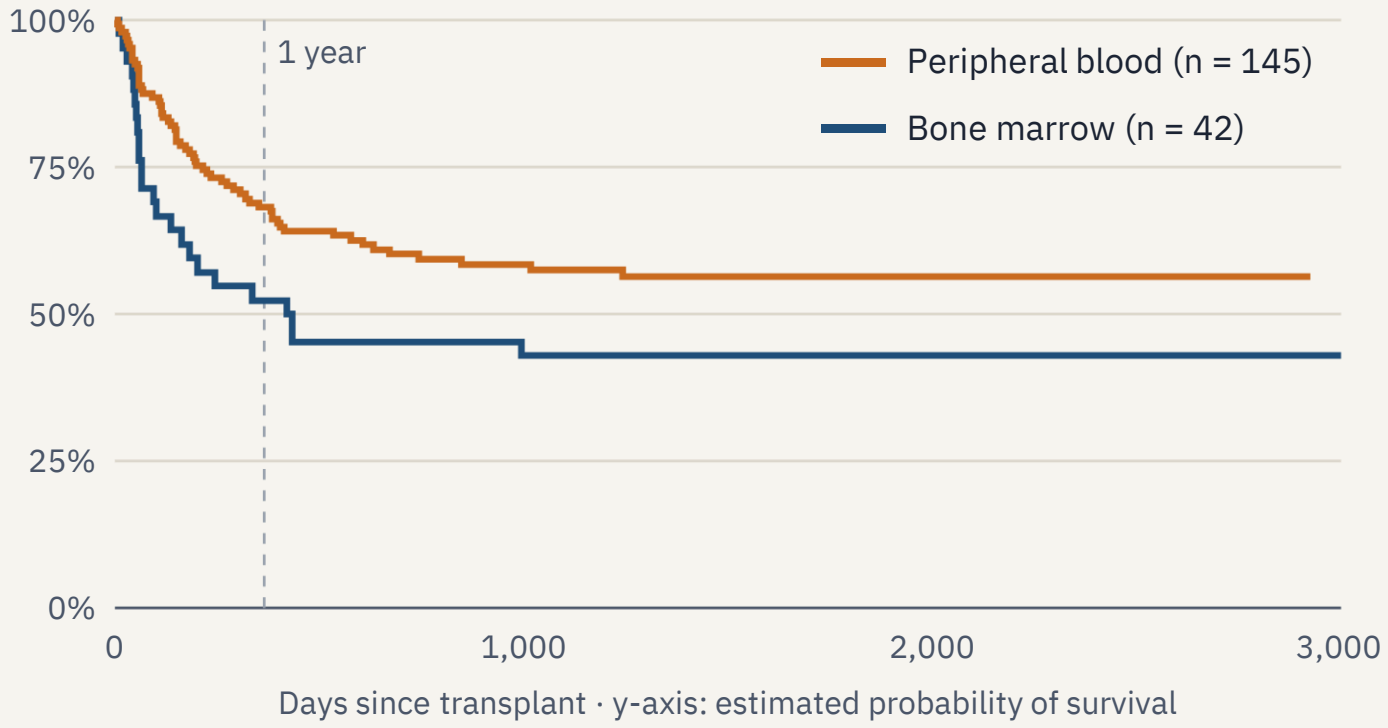
- Aalen–Johansen
- Cause-specific Cox
- Death = competing

Guardrails against overfitting (85 deaths)

At most 9 Cox parameters (≈ 9.4 events each) · forest settings fixed in advance, not tuned · every ML metric measured on held-out folds

KAPLAN–MEIER

PB looks better early, but only at the edge of significance



Survival at 1 year

68% vs 52%

PB vs BM · 5 years: 56% vs 43%

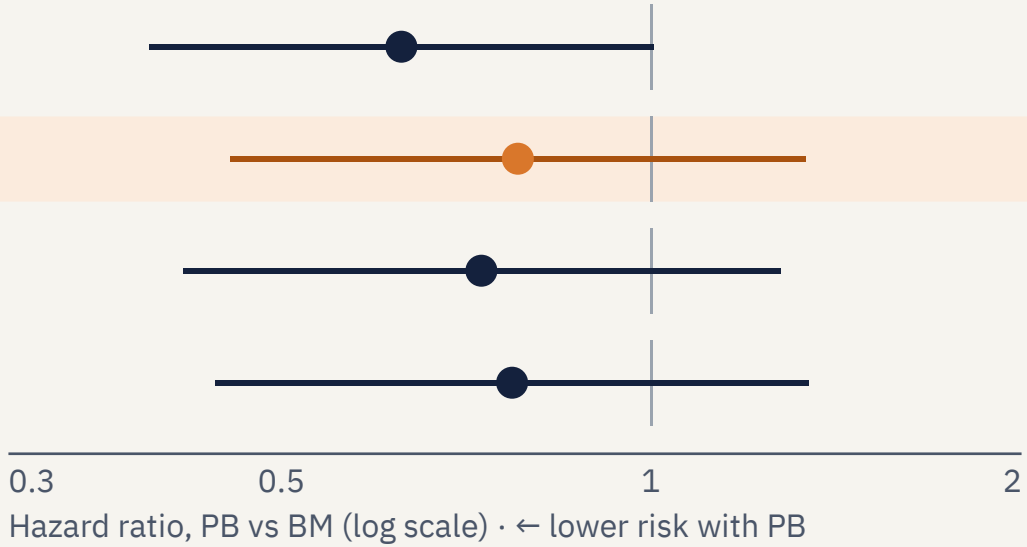
Log-rank test

$\chi^2 = 3.83$, $p = 0.050$

H1a not supported at $\alpha = 0.05$: a borderline result, and unadjusted.

COX MODELS · TESTING H1B

Adding CD34+ dose pulls the source effect toward 1



Model 1 · source only — 0.63 (0.39–1.00), p = 0.052

Model 2 · + CD34+ dose — 0.78 (0.45–1.33), p = 0.36

Model 3 · + age, risk, disease — 0.73 (0.42–1.27), p = 0.27

Model 3q · age as a U-shape — 0.77 (0.44–1.34), p = 0.36

46% of the source log-HR disappears once dose is added. But the bootstrap 95% CI of the change (−0.55 to 0.07) includes 0: **H1b points the right way, not statistically confirmed.**

FINAL COX MODEL (3Q) · TESTING H2

Age and lymphoma matter; source and dose do not

Predictor	HR	95% CI	p
Stem-cell source (PB vs BM)	0.77	0.44–1.34	0.36
CD34+ dose (per 10^6/kg)	0.984	0.952–1.016	0.32
Age, centred (per year)	1.05	1.00–1.10	0.053
Age² (U-shape)	1.011	1.003–1.020	0.010
High-risk group	1.17	0.71–1.92	0.54
AML vs ALL	0.68	0.34–1.39	0.29
Chronic leukaemia vs ALL	0.90	0.50–1.63	0.73
Lymphoma vs ALL (n = 9)	4.20	1.74–10.13	0.001
Non-malignant vs ALL	1.16	0.57–2.35	0.68

H2 not supported: CD34+ HR 0.99, p = 0.57 in Model 3 (0.984, p = 0.32 here)

Fit

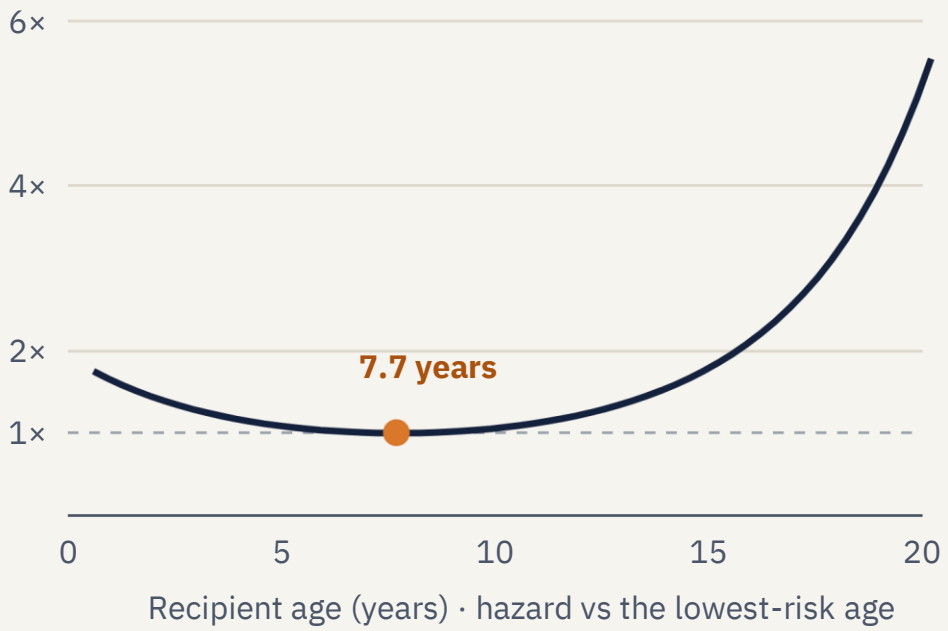
LR test vs linear age: p = 0.013 · AIC 828 vs 832 · C-index 0.64 (in-sample)

Assumptions

No Schoenfeld test p < 0.05 · 9.4 events per parameter

AGE EFFECT AND ROBUSTNESS

Risk is U-shaped in age; the source result is robust



Lowest risk around **7–8 years**; higher in infants and adolescents. Few patients sit at the extremes, so trust the shape more than the exact height.

Source HR in sensitivity analyses

Analysis	HR	p
Main model (3q)	0.77	0.36
Without lymphoma	0.70	0.23
log CD34+ dose	0.82	0.56
Stratified by disease	0.80	0.42

HYPOTHESIS DECISIONS

None of the three hypotheses is confirmed

H1a

Source is associated with survival (unadjusted)

Not supported

Log-rank p = 0.050, borderline

H1b

Dose explains part of the source effect

Right direction only

46% attenuation; bootstrap CI includes 0

H2

Higher dose, longer survival (adjusted)

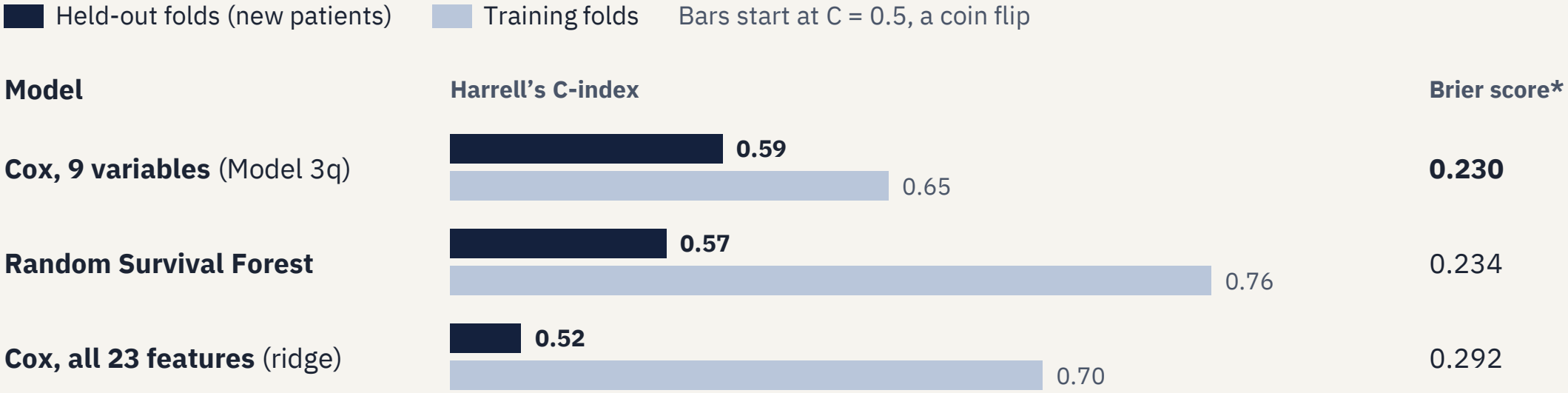
Not supported

HR 0.99 per 10^6/kg, p = 0.57

Not significant is not the same as no effect: with 42 bone-marrow patients and 85 deaths, a true hazard ratio of 0.75 would be detected only about 20% of the time.

MACHINE LEARNING · 5 × 5-FOLD CROSS-VALIDATION

The forest overfits; the simple Cox model generalises best

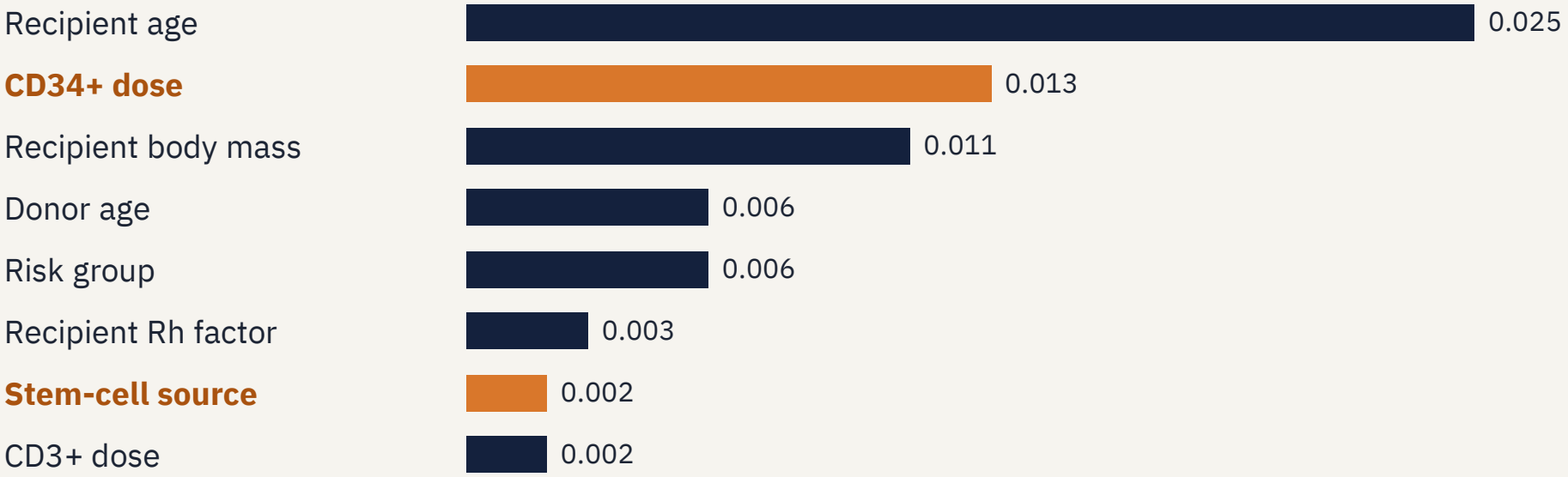


More flexibility, worse on new patients: the forest scores 0.76 on training folds but 0.57 on held-out ones. Baseline data predict survival only modestly.

*Integrated Brier score, days 30–1,500, held-out folds; lower is better, 0.25 = uninformative

RANDOM SURVIVAL FOREST · PERMUTATION IMPORTANCE

Age leads; source adds almost nothing

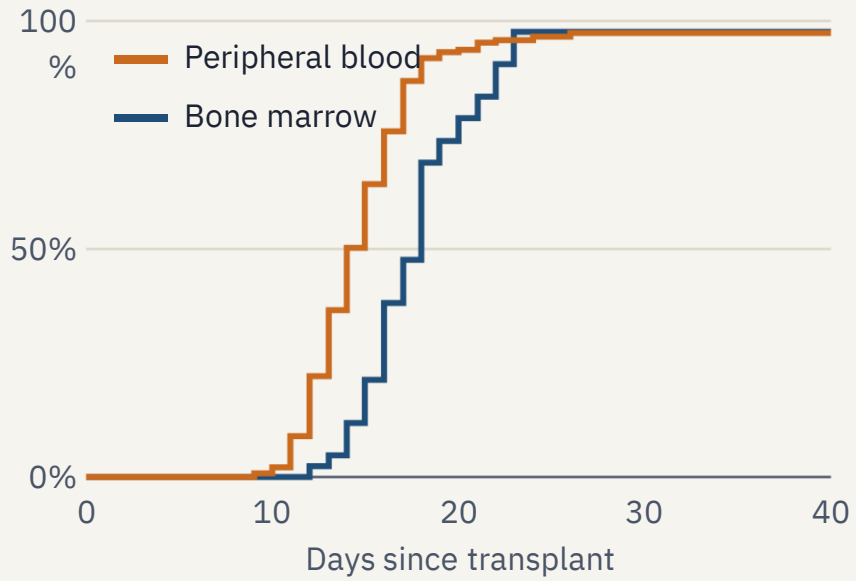


Mean drop in held-out C-index when a variable is shuffled (25 folds). Age, dose and body mass overlap (dose is per kg), so a high rank means **useful for prediction**, not an independent effect.

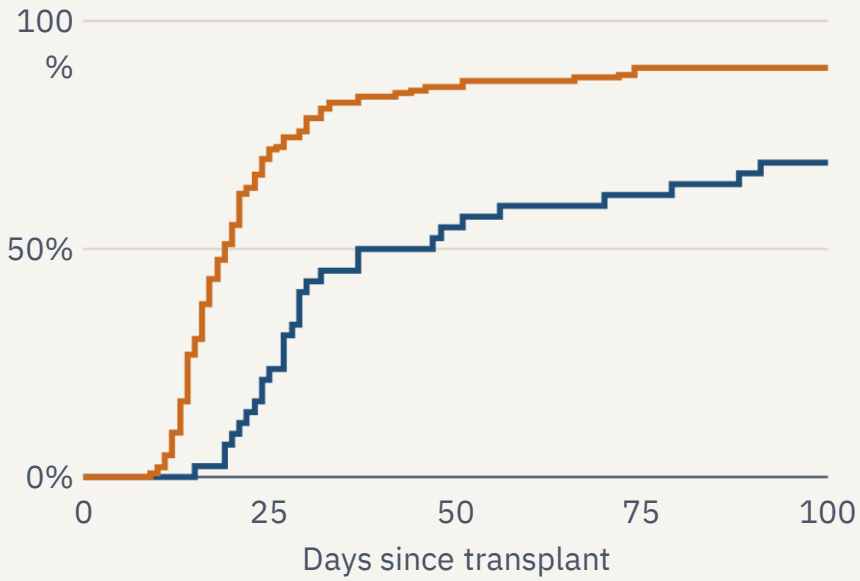
SECONDARY OUTCOME · ENGRAFTMENT

PB grafts engraft faster, and the dose explains why

Neutrophil recovery · median 14 vs 18 days



Platelet recovery · median 18 vs 29 days



Adjusted for dose and age, **CD34+ dose speeds recovery** (neutrophils cs-HR 1.46, p = 0.002; platelets 2.29, p < 0.001) and source is no longer significant (p = 0.16; p = 0.94).

Aalen–Johansen cumulative incidence; death before recovery = competing event

CONCLUSIONS

What we can and cannot say

1. **Source:** PB looked better before adjustment (HR 0.63, p = 0.05). After adjustment HR ≈ 0.75, not significant.

2. **Source and dose are entangled:** PB delivers 3.3× the CD34+ dose; adding dose removes about half the source effect, but not beyond noise.

3. **Dose matters for engraftment, not survival:** faster neutrophil and platelet recovery, no adjusted survival link.

4. **What predicts survival:** age (U-shaped) and lymphoma. Prediction is modest (C ≈ 0.59), and the simple model beats the complex one.

LIMITATIONS AND NEXT STEPS

A small, observational dataset sets the limits

Limitations

- 187 patients, 85 deaths: about 20% power for HR 0.75
- Source not randomised: centre, era and urgency may confound
- Tiny subgroups: lymphoma n = 9, bone marrow n = 42
- Age² chosen after inspecting residuals
- Internal validation only, no external cohort

Next steps

- Replicate in larger registry data
- Formal mediation analysis: source → dose → outcome
- GvHD and relapse as time-dependent covariates
- Separate transplant-related death from relapse death
- External validation of the prediction model

Thank you

Questions?

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